higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T1390**(E)**(A11)T

NATIONAL CERTIFICATE

POWER MACHINES N5

(8190035)

11 April 2019 (X-Paper)
09:00–12:00

REQUIREMENTS: **Steam Tables (BOE 173)**
Superheated steam tables (Appendix to BOE 173)

Students need drawing instruments.

Non-programmable calculators may be used.

This question paper consists of 5 pages and a formula sheet of 3 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
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NATIONAL CERTIFICATE
POWER MACHINES N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1

1.1 There are two laws of thermodynamics, namely Joule's law (the first law) and Clausius' theory (the second law).

State Clausius's theory. (2)

1.2 What is the function of a governor in a mechanical mechanism? (2)

1.3 What are the TWO basic products of combustion resulting from a chemical reaction between a hydrocarbon and oxygen? (2)

1.4 Name TWO particulars of manufacturers' information that should be engraved or printed on the boiler. (2)

1.5 Name the TWO classifications of steam turbines depending on how steam expansion takes place. (2)

[10]

QUESTION 2

A certain gas with a volume of 0,5 m^3 is heated isothermally from an initial pressure of 220 kPa and a temperature of 30 °C to a pressure of 1,5 MPa. The gas is then further expanded at a constant volume to a temperature of 20 °C. The gas characteristic constant is 0,296 kJ/kg. K and the specific heat capacity at constant volume is 0,709 kJ/kg.K.

Calculate the following:

2.1 The volume at the end of the isothermal process (3)

2.2 The final pressure at the end of the constant volume process (3)

2.3 The specific heat capacity at constant pressure in kJ/kg.K (2)

2.4 The quantity of heat transferred during the isothermal process and the heat received (absorbed) during the expansion process in kJ/kg (4)

2.5 The total change in internal energy for the entire process in kJ/kg (4)

2.6 The total change in entropy for the entire process in kJ/kg (4)

[20]

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QUESTION 3

Steam with a mass of 2 kg at 2 500 kPa and 250 °C is expanded to 250 kPa. The steam is 9% wet. The index of expansion $n = 1{,}3$. The specific heat capacity of superheated steam is 2 kJ/kg.°C.

Calculate the following:

3.1 The total enthalpy of superheated steam in kJ (4)

3.2 The total volume of superheated steam in m^3 (4)

3.3 The total enthalpy of wet steam in kJ (4)

3.4 The total volume of wet steam in m^3 (4)

3.5 The total change in internal energy in kJ (4)

[20]

QUESTION 4

4.1 A quantity of 3 200 kg of steam per hour flows into a condenser at a pressure of 11 kN/m^2 and is 9% wet. The air leakage is 1,5 kg per 1 000 kg steam. The temperature of the air pump and the condensate is 41,6 °C. Cooling water has a mass flow rate of 12 kg/s and it enters at a temperature of 20 °C. Take the gas characteristic constant as 0,296 kJ/kg.K and the specific heat capacity of water as 4,187 kJ/kg.°C.

Calculate the following:

4.1.1 The outlet (final) temperature of the cooling water (6)

4.1.2 The capacity of the air pump in m^3/minute (5)

4.2 The composition of a gaseous fuel has the following structural analysis according to volume:

H_2 = 37%
CH_4 = 29%
CO = 18%
CO_2 = 1%
N_2 = 15%

Air composition consists of 21% oxygen and 79% nitrogen

Calculate the following:

4.2.1 The volume of air required to completely burn 1 m^3 of the fuel (5)

4.2.2 The total percentage analysis, by volume, of ALL the products of combustion (4)

[20]

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QUESTION 5

In a steam turbine test rig, a recorded quantity of 2 kg of steam per hour enters an impulse steam turbine at a velocity of 900 m/s and is discharged through a nozzle inclined at 20° to plane of rotation of the turbine. The steam enters a series of turbine runner blades that are rotating at an average tangential velocity of 350 m/s, with each runner blade having an outlet angle of 32°. Take the relative velocity coefficient for the turbine system as the steam flows over the runner blades as 0,75.

5.1 Construct a velocity diagram in the ANSWER BOOK on landscape to a scale of 1 cm = 50 m/s. Indicate all the angles and the velocity values in m/s on the velocity diagram.

NO MARKS WILL BE AWARDED FOR THE VELOCITY DIAGRAM IF THE INSTRUCTIONS ARE NOT FOLLOWED ACCORDINGLY. (10)

5.2 Use the diagram to determine the following:

5.2.1 The angle at which the steam leaves the turbine

5.2.2 The inlet angle of the moving blade

(2 × 1) (2)

5.3 Use the diagram to calculate the power developed by the turbine. (3)
[15]

QUESTION 6

Air enters a single stage, single-acting reciprocating air compressor at a rate of 15 m^3 per minute at a temperature of 42 °C and a pressure of 100 kPa. The air leaves the compressor at a pressure of 700 kPa. The compressor runs at a speed of 350 r/min and a clearance volume of 5% of the swept volume. Take the index of compression as n = 1,3 and the mechanical efficiency as 85% of the reciprocating compressor.

Calculate the following:

6.1 The cylinder volume of the compressor in m^3 (9)

6.2 The delivery temperature of the compressor (2)

6.3 The power required to drive the compressor in kW (4)
[15]

TOTAL: 100

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POWER MACHINES N5

FORMULA SHEET

1. $Q = W + \Delta U$
2. $\Delta U = mC_{\upsilon}\Delta T$
3. $Q = mC_{\upsilon}\Delta T$
4. $Q = mC_{p}\Delta T$
5. $Q = P_1V_1 \ln \frac{V_2}{V_1}$
6. $\Delta S = m\left(C_{\upsilon} \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1} \right)$
7. $W = P_1\Delta V$
8. $W = P_1V_1 \ln \frac{V_2}{V_1}$
9. $W = \frac{P_1V_1 - P_2V_2}{n-1}$
10. $W = \frac{P_1V_1 - P_2V_2}{\gamma - 1}$
11. $R = C_p - C_{\upsilon}$
12. $\gamma = \frac{C_p}{C_{\upsilon}}$
13. $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$
14. $PV = mRT$
15. $P_1V_1 = P_2V_2$
16. $P_1V_1^n = P_2V_2^n$
17. $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} = \left(\frac{V_1}{V_2}\right)^{n-1}$
18. $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$
19. $h = h_f + \chi h_{fg}$
20. $h = h_g + C_p\Delta T$
21. $h = h_f + h_{fg} = h_g$
22. $V_{\text{sup}} = \frac{n-1}{n}\left(\frac{h_{\text{sup}} - 1\,941}{P}\right)$
23. $\chi = \frac{V_m}{V_g}$
24. $\chi = \frac{M}{M + m}$
25. $U = H - PV$
26. $gZ_1 + U_1 + P_1V_1 + \frac{1}{2}C_1^2 + Q =$

    $gZ_2 + U_2 + P_2V_2 + \frac{1}{2}C_2{}^2 + W$
27. $\eta = \frac{\dot{m}_s(h_2 - h_1)}{\dot{m}_f\, CV}$
28. $EE = \frac{\dot{m}_s(h_2 - h_1)}{\dot{m}_f\ 2\,257}$
29. $p = (B_m \pm M_m)\frac{101{,}325}{760}$

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30. $m = \frac{100}{23}\left[C\frac{8}{3} + 8H_2 + S - O_2\right]$

31. $C_xH_y + \left(x + \frac{y}{4}\right)O_2 = xCO_2 + \frac{y}{2}H_2O$

32. $H.C.V. = (CV_C.C) + CV_{H_2}\left(H_2 - \frac{O_2}{8}\right) + (CV_s.S)$

33. $L.C.V. = H.C.V. - h_{fg}\,(9H_2)$

34. $H.C.V. = \frac{(m_w + m_e)\,C_p\Delta T}{m_f}$

35. $W = P_1V_e\left(\frac{n}{n-1}\right)\left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right] = mRT_1\left(\frac{n}{n-1}\right)\left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right]$

36. $\eta_c = \frac{V_e}{V_s}.100 = 1 - \frac{Vc}{Vs}\left[\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}} - 1\right] = 1 + \alpha - \alpha(r_p)^{\frac{1}{n}}$

37. $\eta_\alpha = \frac{V_\alpha}{V_s}.100$

38. $F_c = \dot{m}(C_{fe} - C_{fi})$

39. $P = \dot{m}U[C_{wi} - (-C_{we})]$

40. $\eta = \frac{2U[C_{wi} - (-C_{we})]}{C_{ai}{}^2}.100$

41. $U = \pi DN$

42. $\dot{m}V = AC$

43. $(m + M)\,g = m\omega^2 h$

44. $Vs = \frac{\pi}{4}D^2L$

45. $\theta_1 = t_c - twi$

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46. $\theta_2 = t_c - two$

47. $$Log.temp.diff. = \frac{\theta_1 - \theta_2}{\ell n \frac{\theta_1}{\theta_2}}$$

48. $$Piso = P_1 V_1 \ell n \left(\frac{P_2}{P_1} \right)$$

49. $$Pact = \frac{n}{n-1} P_1 V_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

50. $$Niso = \frac{Piso}{Pact} \bullet 100$$

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